

SYSTEMATIC REVIEW

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Artificial intelligence in predicting recurrence after first-line treatment of liver cancer: a systematic review and meta-analysis

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Abstract

Background The aim of this study was to conduct a systematic review and meta-analysis to comprehensively evaluate the performance and methodological quality of artificial intelligence (AI) in predicting recurrence after single first-line treatment for liver cancer.

Methods A rigorous and systematic evaluation was conducted on the AI studies related to recurrence after single first-line treatment for liver cancer, retrieved from the PubMed, Embase, Web of Science, Cochrane Library, and CNKI databases. The area under the curve (AUC), sensitivity (SENC), and specificity (SPEC) of each study were extracted for meta-analysis.

Results Six percutaneous ablation (PA) studies, 16 surgical resection (SR) studies, and 5 transarterial chemoembolization (TACE) studies were included in the meta-analysis for predicting recurrence after hepatocellular carcinoma (HCC) treatment, respectively. Four SR studies and 2 PA studies were included in the meta-analysis for recurrence after intrahepatic cholangiocarcinoma (ICC) and colorectal cancer liver metastasis (CRLM) treatment. The pooled SENC, SEPC, and AUC of AI in predicting recurrence after primary HCC treatment via PA, SR, and TACE were 0.78, 0.90, and 0.92; 0.81, 0.77, and 0.86; and 0.73, 0.79, and 0.79, respectively. The values for ICC treated with SR and CRLM treated with PA were 0.85, 0.71, 0.86 and 0.69, 0.63, 0.74, respectively.

Conclusion This systematic review and meta-analysis demonstrates the comprehensive application value of AI in predicting recurrence after a single first-line treatment of liver cancer, with satisfactory results, indicating the clinical translation potential of AI in predicting recurrence after liver cancer treatment.

Keywords Liver cancer, Artificial intelligence, Recurrence, Percutaneous ablation, Surgical resection, Transarterial chemoembolization

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Introduction

Liver cancer poses a significant global public health burden, especially in China, which accounts for nearly half of liver cancer burden. Liver cancer is classified into primary and secondary liver cancer. Primary liver cancer originates in the liver itself, with pathological types including hepatocellular carcinoma (HCC), intrahepatic cholangiocarcinoma (ICC), and mixed hepatocellular-cholangiocarcinoma. HCC comprises 80% of the global liver cancer burden, followed by ICC at 15% [1]. Secondary liver cancer, on the other hand, arises from the spread of cancer originating from non-hepatic tissues to the liver. According to research, among 2.4 million cancer patients, 5.14% experienced liver metastasis, and the most common primary site for young women was breast cancer, while for young men, it was colorectal cancer [2]. The first-line treatment options for liver cancer include liver surgical resection (SR), percutaneous ablation (PA), and transarterial chemoembolization (TACE). For HCC, the risks and benefits of first-line treatment options must be considered, and personalized treatment should be provided based on tumor factors and the patient's functional status. SR remained the preferred first-line treatment for HCC, particularly in Barcelona Clinic Liver Cancer stages 0-A. Although PA was characterized by minimal invasiveness and rapid recovery, meta-analyses of randomized controlled trials had shown that SR offered longer disease-free survival and a lower recurrence rate compared to PA [3]. However, PA was considered the optimal treatment option for early-stage HCC that was not amenable to resection [4]. TACE was often used as a bridging therapy, in combination with SR or PA, to maximize patient benefits [5]. SR was considered the only viable and potentially curable first-line treatment for ICC, but resectability and curability remained low [6]. Lymph node metastasis was the primary cause of poor long-term survival after SR in ICC patients, and portal lymph node dissection was a common surgical approach for staging. However, for unresectable ICC, PA as an alternative therapy had shown to improve survival rates in patients with ICC tumors smaller than 3 cm, similar to SR [7]. TACE was used exclusively for palliative treatment of ICC to improve patients' quality of life. As for colorectal liver metastases (CRLM), regardless of the systemic therapy administered, SR remained the gold standard for curing rectal cancer with CRLM, but only 20% of patients with solitary CRLM had the opportunity to undergo SR [8]. While surgical resection was considered the preferred option, based on current literature, there was no evidence to suggest that non-surgical therapies were inferior to SR. For instance, a retrospective study found that PA was suitable for elderly males with CRLM, particularly those with tumors smaller than 2 cm,

compared to SR [9]. Despite the widespread application of these three first-line treatment options in liver cancer, the high recurrence rate remains a significant challenge. For example, a single-center study revealed that among 1235 HCC patients who underwent SR and 517 HCC patients who received PA, the early recurrence (ER) rates were 19% and 37%, respectively [10]. In a prospective study, 259 ICC patients treated with SR had an ER rate of 50% during follow-up, with a late recurrence rate of 29% [11]. Additionally, a multicenter study showed that among 207 patients with CRLM who underwent PA, the ER rate was 48% [12]. Given these findings, effectively predicting the recurrence of liver cancer after first-line treatment has become a widely explored area of studies.

Accurately predicting the recurrence of liver cancer after treatment has profound implications for treatment planning and long-term management strategies. To achieve this, various predictive models and biomarkers have been extensively explored as they offer insights into recurrence risk. The advancement of such predictive methods has the potential to ultimately lead to more personalized treatment plans and improved outcomes for cancer patients. For instance, a multicenter prospective study constructed a predictive model utilizing factors such as tumor diameter, tumor number, large vessel invasion, satellite nodules, alpha-fetoprotein, HBV-DNA level, and prothrombin time. This model was thoroughly validated through internal, external, and prospective cohorts, accurately predicting ER after HCC-SR [13]. Furthermore, a machine learning model developed based on 14 clinicopathological features of 536 ICC patients had been used to predict ER after ICC-SR, aiming to provide personalized consultation, treatment, and advice [14]. Additionally, a nomogram for predicting ER after CRLM-SR was formulated using factors like the number of liver metastases, suspected lymph node involvement, neurovascular invasion, colon/rectum location, albumin levels, post-operative carcinoembryonic antigen (CEA), and carbohydrate antigen 19-9 (CA19-9) levels. This nomogram achieved area under the curve (AUC) values of 0.87 and 0.79 in internal and external validation, assisting in decision-making for selective follow-up strategies [15]. Moreover, several meta-analyses have shed light on biomarkers associated with ER after first-line treatment for liver cancer from a large sample perspective. For example, a meta-analysis encompassing nine ICC-ER-related studies revealed that preoperative factors such as cirrhosis, gender, and HBV status were not significantly associated with ER. However, among postoperative pathological factors, multiple tumors, microvascular invasion, large vessel invasion, lymph node metastasis, and R1 resection were identified as risk factors for increased ER [16]. Similarly, a meta-analysis of 24 CRLM-ER-related

studies indicated that elevated levels of CEA and CA199, larger and bilobar liver metastases, extensive liver resection, positive surgical margins, and postoperative complications were associated with an increased risk of ER [17]. It is crucial to note, however, that these analyses are primarily based on observational studies investigating clinical and pathological factors related to ER following first-line treatment. Furthermore, some of them are analyzed from a postoperative perspective. Observational studies, due to the interference of numerous confounding factors and the lack of balanced baseline characteristics between cohorts, may not always yield reliable conclusions. Therefore, future research with rigorous experimental designs and comprehensive analysis is warranted to enhance the accuracy and reliability of predictive models for liver cancer recurrence.

Artificial intelligence (AI) has been elucidated as an alternative biomarker for predicting cancer recurrence after treatment, demonstrating satisfactory results compared to clinical pathological features. AI utilizes computer technology to extract deeper-level information from medical images such as computed tomography (CT) [18], magnetic resonance imaging (MRI) [19], or ultrasound (US) [20], encompassing radiomics and deep learning [21]. For instance, a machine learning radiomics model developed based on contrast-enhanced ultrasound (CEUS) images of 318 HCC patients had been validated in a test cohort to predict ER after PA, which radiomics model outperformed the clinical model [22]. Similarly, a radiomics model developed from CT images of 138 ICC patients also showed superior predictive capabilities for ER after SR compared to clinical models [23]. However, upon further investigation, it had been observed that only AI meta-analyses focusing on HCC treatment recurrence were available, and these had not conducted independent analyses for various first-line treatment protocols. Moreover, there was a lack of comprehensive meta-analyses evaluating the recurrence after first-line treatment in ICC and CRLM. The potential of AI in predicting cancer recurrence is immense, yet its full application and understanding remain to be explored.

Therefore, this study aimed to conduct a systematic review and meta-analysis to analyze the application of AI in predicting recurrence after first-line treatment for liver cancer, including HCC, ICC and CRLM. The goal was to clarify the current study status of AI in recurrence prediction, as well as to evaluate the methodological quality of the studies. By systematically reviewing and synthesizing the available evidence, we hope to provide a comprehensive understanding of the effectiveness and limitations of AI in predicting cancer recurrence, thereby guiding future research and clinical

applications. We present this article in accordance with the PRISMA reporting checklist (Supplementary materials 1-PRISMA_2020_checklist).

Methods

Protocol and registration

The review protocol was registered on the Prospective Register of Systematic Reviews (PROSPERO; <https://www.crd.york.ac.uk>; registration number CRD42024558716).

Study search strategy

A comprehensive search was undertaken in the PubMed, Embase, Web of Science, Cochrane Library, and CNKI databases before May 10, 2024, utilizing the following keywords: “radiomics”, “deep learning”, “liver”, “ablation”, “resection”, “TACE”, “transarterial chemoembolization”, “HCC”, “hepatocellular carcinoma”, “ICC”, “cholangiocarcinoma”, “metastasis”, and “metastases”. The objective was to retrieve studies pertaining to the utilization of AI in forecasting recurrence post-primary treatment of focal liver lesions. Following the initial search, all relevant literature concerning recurrence was thoroughly scrutinized, primarily through the examination of abstracts and the full text. The inclusion process was independently evaluated by two reviewers, each with over three years of expertise in abdominal imaging diagnosis. Any inconsistencies were resolved by a senior reviewer, boasting over five years of experience in abdominal imaging diagnosis.

Study selection criteria

The studies must adhere to the following criteria: (1) study type: original study; (2) population: patients with liver cancer, including HCC, ICC, and CRLM; (3) treatment strategy: receiving only a single first-line treatment, including SR, PA, and TACE; (4) the pre-treatment imaging examinations include US, CT, or MRI; (5) objective: quantitative analysis of pre-treatment images using radiomics and deep learning AI methods; (6) results: sufficient data to reconstruct a 2×2 sensitivity (SENC) and specificity (SPEC) table for predicting post-treatment recurrence.

The studies were excluded if they met the following criteria: (1) any comments, meta-analyses, case reports, guidelines, errata, or duplicate cohort studies; (2) the use of combined treatment strategies; (3) AI quantitative analysis using post-treatment images or images of non liver cancer; (4) the presence of anti-tumor treatments such as radiotherapy, chemotherapy, or targeted therapy before first-line treatment; (5) controversial predictive purposes; (6) insufficient data extraction.

Data extraction

The extraction of study data was independently conducted by a reviewer with 3 years of study experience and independently verified by another reviewer with the same experience. The extraction encompassed information such as the first author, publication year, country of publication, AI methods (radiomics or deep learning), pathological types of liver cancer, design type (prospective or retrospective), number of centers (single or multi-center), image type (US, CT, or MRI), follow-up strategy (within 2 years or longer), and a 2×2 SENC and SEPC table comprising true positives (TP), false negatives (FN), false positives (FP), and true negatives (TN). In cases where two or more AI models were proposed based on the same cohort, the model with higher diagnostic accuracy was prioritized for analysis.

Quality assessment

The Radiomics Quality Score (RQS), Checklist for AI in Medical Imaging (CLAIM), and Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) were applied to evaluate the methodological quality of the included studies [24]. RQS included 16 evaluation indexes, including image acquisition, image preprocessing, validation, performance evaluation, practicality, open science. CLAIM included 42 evaluation indexes, compared to RQS, a further subdivision and evaluation of all steps related to AI was conducted. QUADAS-2 was used to

evaluate the risk of bias at the study level, including bias in fields such as: (1) patient selection, (2) index test, (3) reference standard, and (4) flow and timing. A modified version of QUADAS-2 proposed by Sollini et al. and verified by Bedrikovetski S et al. was used [25].

Statistical analysis

Meta-analysis of TP, TN, FP, and FN was performed using the “midas” module in Stata (version 12.0) and Meta-DiSc (version 1.4), and the pooled SENC and SEPC were determined based on the fixed-effects model. The summary receiver operating characteristic (sROC) curve assessed the value of AI in predicting recurrence after first-line treatment. Heterogeneity among studies was evaluated using the Cochrane Q-test (p -value < 0.05) and I^2 statistic (> 50%). The Deeks’ funnel plot asymmetry test was employed to assess publication bias. All statistical significance was set at a two-sided p -value < 0.05.

Results

Results of search strategy

The results of the search strategy were shown in Fig. 1. A total of 98 AI studies related to PA, 327 studies related to SR, and 75 studies related to TACE were identified and rigorously evaluated. After initial screening, 69 PA-related studies, 240 SR-related studies, and 52 TACE-related studies were excluded due to their irrelevance to recurrence, leaving 29 PA-related studies, 89 SR-related

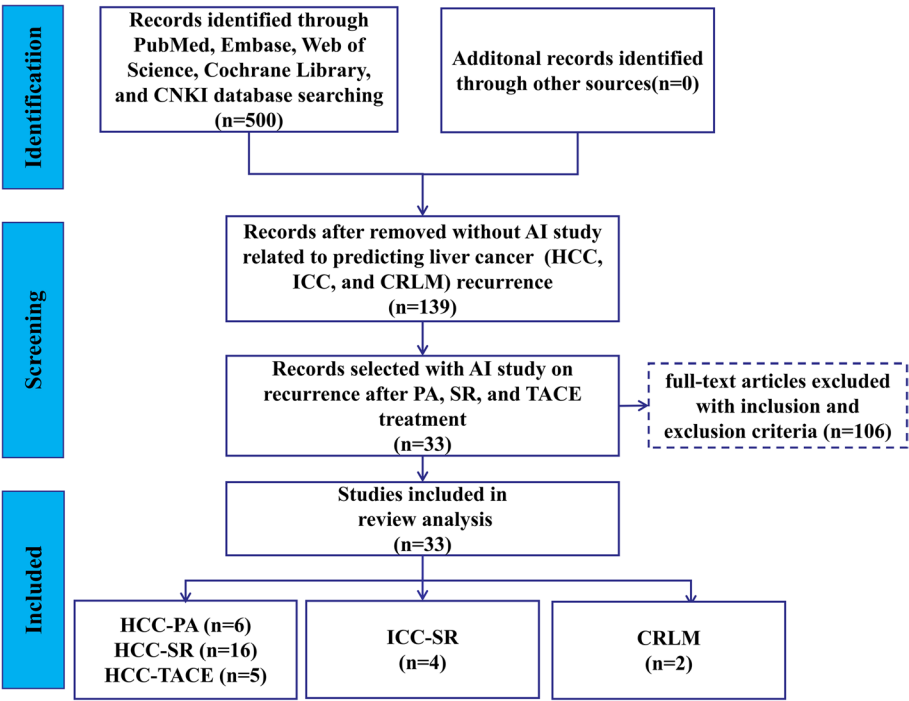


Fig. 1 Search strategy for study inclusion

studies, and 23 TACE-related studies for further analysis. Among the 123 studies, 80 studies were excluded because they were unable to extract the data of SENC and SPEC or the number of recurrent cases was unknown, 8 studies adopted two kinds of first-line treatments [26–33], and 5 studies did not conduct model validation [34–38]. In the SR-related studies, two came from the same institution, and the one with lower performance [39] was excluded.

After these exclusions, in the application of AI in predicting HCC recurrence, a final total of 6 PA-related studies [22, 40–44], 16 SR-related studies [45–60], and 5 TACE-related studies [61–65] were included in this study.

In the application of AI in predicting ICC recurrence, no studies had been found on the application after PA or TACE for ICC, and only 11 studies related to SR had been identified [23, 66–75], with 4 of them finally included in the meta-analysis [23, 67, 73, 75].

In the application of AI for predicting the recurrence of metastases, most of them were of the CLRM pathological type, therefore, only AI for CLRM were analyzed. In the application of AI in predicting CLRM recurrence, only 8 were related to PA [12, 33, 36, 76–80], 9 were related to SR [81–89], and 1 was related to TACE [90]. After exclusion, only 2 PA-related studies [12, 80], 1 SR-related study [81], and 0 TACE-related studies met the criteria for the meta-analysis.

Characteristics of the included studies

Except for one study that adopted a prospective design, the remaining studies included in the meta-analysis all employed a retrospective design. Only 8 studies utilized multi-center data, and 6 studies performed external data validation. According to the affiliation of the first author, only one study originated from the United States, while the remaining studies were all from China. All studies defined recurrence, including progression of the primary lesion, intrahepatic recurrence, and extrahepatic metastasis. Regarding the imaging protocols, with 17 using a single scanner, 8 using multiple scanners, and 8 with unclear scanner details. 4 studies were based on US images, 14 on CT images, and 15 on MRI images. 29 studies relied on manual segmentation by multiple individuals, 19 used ITK-SNAP software, but only 25 performed an intraclass correlation coefficient analysis, and 2 did not report segmentation details. The segmentation methods were manual in 30 studies and semi-automatic in only 3 studies. In terms of image preprocessing, only 16 studies performed resampling. Over half of the studies extracted features based on the Pyradiomics software package. 31 studies employed a multi-step feature reduction strategy, while 2 study was unclear. 22 studies used cross-validation for performance assessment, and 20

studies compared their performance with other methods. 29 studies performed multivariate analysis, and 31 studies adopted ensemble modeling strategies. 23 studies conducted calibration and 22 studies conducted decision curve analyses (Table 1).

Quality assessment

Based on the analysis of the aforementioned characteristics, the RQS, QUADAS-2, and CLAIM methods were used to assess the methodological quality of the studies. The RQS results showed a median total score of 16, ranging from 10 to 22. In the application of HCC, among different first-line treatment strategies such as PA, SR, and TACE, the median RQS scores were 12, 16, and 18, with ranges of 10–16, 10–22, and 12–18, respectively. In the application of ICC, the median RQS scores of SR was 16, with ranges of 10–16. In the application of CLRM, the median RQS scores of PA was 14.5, with ranges of 14–15. None of the studies reported cost-effectiveness, phantom studies, the development of science (Supplementary materials 2-RQS).

The median CLAIM score was 71% (30/42). Some CLAIM items were typically reported, including abstract structure, study introduction, study goal, inclusion criteria for the study population, model performance metrics, performance metrics to distinguish the optimal model(s), and study limitations. However, there were also some items that were often not reported, such as the selection of data subsets, de-identification methods, handling of missing data, rationale for choosing the reference standard, anticipated sample size and how it was determined, failure analysis of incorrectly classified cases, and availability of the full study protocol (Supplementary materials 3-CLAIM).

QUADAS-2 was used to analyze the risk of bias and applicability questions. 29 studies were deemed to have a low risk of bias. However, bias risks and application issues related to the index test were frequently observed, primarily due to the lack of detailed reporting on image preprocessing. Regarding the study flow and timing, the risk of bias remained unclear across all 33 studies (Supplementary materials 4-QUADAS-2). The quality assessment was displayed in Fig. 2.

Comprehensive performance of recurrence prediction after first-line treatment of HCC based on artificial intelligence

The baseline characteristics in the meta-analysis of HCC recurrence prediction were presented in Table 2. One of the PA studies and one of the SR studies only entered a meta-analysis of combined models [41, 58]. A total of 57 cohorts with 5,772 patients were analyzed, including 13 cohorts with 1245 patients treated by PA,

Table 1 Characteristics of 33 studies

Author	Country	Type	Center	Pathology	Treatment	AI	Follow up	Image protocol	Image	Segmentation	Extraction tool	Modeling
Ma QP et al. (2021)	C	R	S	HCC	PA	RAD	2 year	Yes	CEUS	NA	PyRadiomics	MLP
Lv X et al. (2021)	C	R	S	HCC	PA	RAD	0.5 year	Yes	MRI	ITK-SNAP	AK	LASSO
Yang X et al. (2022)	C	R	S	HCC	PA	RAD	1 year	Yes	MRI	ITK-SNAP	LIFEx	COX-LASSO
Chen Cet al. (2023)	C	R	S	HCC	PA	DLR	0.5 year	Yes	MRI	MATLAB	PyRadiomics, deep learning (ResNet18)	SVM
Li M et al. (2022)	C	R	S	HCC	PA	RAD	2 year	Yes	CEUS	MRICroGL	PyRadiomics	SVM-RFE
Li W et al. (2022)	C	R	S	HCC	PA	RAD	1 year	Yes	MRI	ITK-SNAP	United Imaging AI System	RF
Gao W et al. (2022)	C	R	S	HCC	SR	DLR	2 year	Yes	MRI	ITK-SNAP	Deep learning (CNN), Pyradiomics	LASSO
Kang W et al. (2023)	C	R	S	HCC	SR	RAD	2 year	Yes	CT	ITK-SNAP	AK	LASSO
Wang J et al. (2022)	C	R	M	HCC	SR	DL	2 year	Yes	CT	ITK-SNAP	Deep learning (U-Net)	LR
Wang Q et al. (2023)	C	R	S	HCC	SR	RAD	2 year	Yes	MRI	ITK-SNAP	PyRadiomics	SVM
Zhao Y et al. (2021)	C	R	S	HCC	SR	RAD	2 year	Yes	MRI	ITK-SNAP	AK	LR
Zhang Z et al. (2019)	C	P	S	HCC	SR	RAD	1 year	Yes	MRI	ITK-SNAP	AK	LASSO
Zhang H et al. (2022)	C	R	S	HCC	SR	DLR	1 year	Yes	CEUS	Mazda	Mazda, deep learning (ResNet-50)	RF, SVM
Yan M et al. (2023)	C	R	M	HCC	SR	DL	2 year	Yes	MRI	MITK	MITK,Deep learning (VGGNet-19)	GP
Wu C et al. (2022)	C	R	S	HCC	SR	RAD	1 year	Yes	CT	ITK-SNAP	AK	LR
Wang L et al. (2022)	C	R	S	HCC	SR	RAD	2 year	Yes	MRI	ITK-SNAP	AK	LASSO
Ning Pet al. (2020)	C	R	S	HCC	SR	RAD	1 year	Yes	CT	ITK-SNAP	MATLAB	LASSO
Lv C et al. (2023)	C	R	S	HCC	SR	DLR	3 year	Yes	CT	ITK-SNAP	Pyradiomics, deep learning (ResNet-50)	RF
Li W et al. (2022)	C	R	S	HCC	SR	RAD	2 year	Yes	MRI	ITK-SNAP	Pyradiomics	COX
Li J et al. (2024)	C	R	S	HCC	SR	RAD	2 year	Yes	MRI	ITK-SNAP	AK	LR
Li Z et al. (2023)	C	R	S	HCC	SR	RAD	2 year	Yes	CT	3D-SLICER	Pyradiomics	LASSO
Wang L et al. (2023)	C	R	S	HCC	SR	RAD	1 year	Yes	CT	ITK-SNAP	NA	LASSO
Chen C et al. (2023)	C	R	S	HCC	TACE	RAD	1 year	Yes	CT	3D-SLICER	PyRadiomics	LASSO
Peng G et al. (2024)	C	R	S	HCC	TACE	RAD	NA	Yes	MRI	3D-SLICER	PyRadiomics	XGBoost
Jin Z et al. (2021)	C	R	M	HCC	TACE	RAD	NA	Yes	CT	3D-SLICER	PyRadiomics	LDA
Wang DD et al. (2022)	C	R	M	HCC	TACE	RAD	NA	No	CT	MITK	PyRadiomics	XGBoost
Kong X et al. (2021)	C	R	S	HCC	TACE	RAD	0.5 year	Yes	MRI	AK	AK	LR
Jollissaint JS et al. (2022)	US	R	S	ICC	SR	RAD	1 year	Yes	CT	Scout Liver	Matlab	RF
Zhu Y et al. (2021)	C	R	S	ICC	SR	RAD	0.5 year	Yes	CT	NA	Pyradiomics	LASSO
Song Y et al. (2023)	C	R	M	ICC	SR	RAD	1 year	No	CT	3D-SLICER	Pyradiomics	LightGBM
Liang W et al. (2018)	C	R	S	ICC	SR	RAD	1 year	Yes	MRI	ITK-SNAP	MATLAB	LASSO
Zhao QX et al. (2023)	C	R	M	CRIM	PA	DL	2 year	Yes	CEUS	ITK-SNAP	Deep learning (CNN)	MLP
Hu H et al. (2023)	C	R	S	CRIM	PA	RAD	5 year	No	CT	ITK-SNAP	Pyradiomics	SVM

C China, US United States, R retrospective, P prospective, S single-center, M multi-center, HCC hepatocellular carcinoma, ICC intrahepatic cholangiocarcinoma, CRLM colorectal cancer liver metastasis, PA percutaneous ablation, SR surgical resection, TACE transarterial chemoembolization, RAD radiomics, DL deep learning, DLR deep learning radiomics, NA represented as unclear

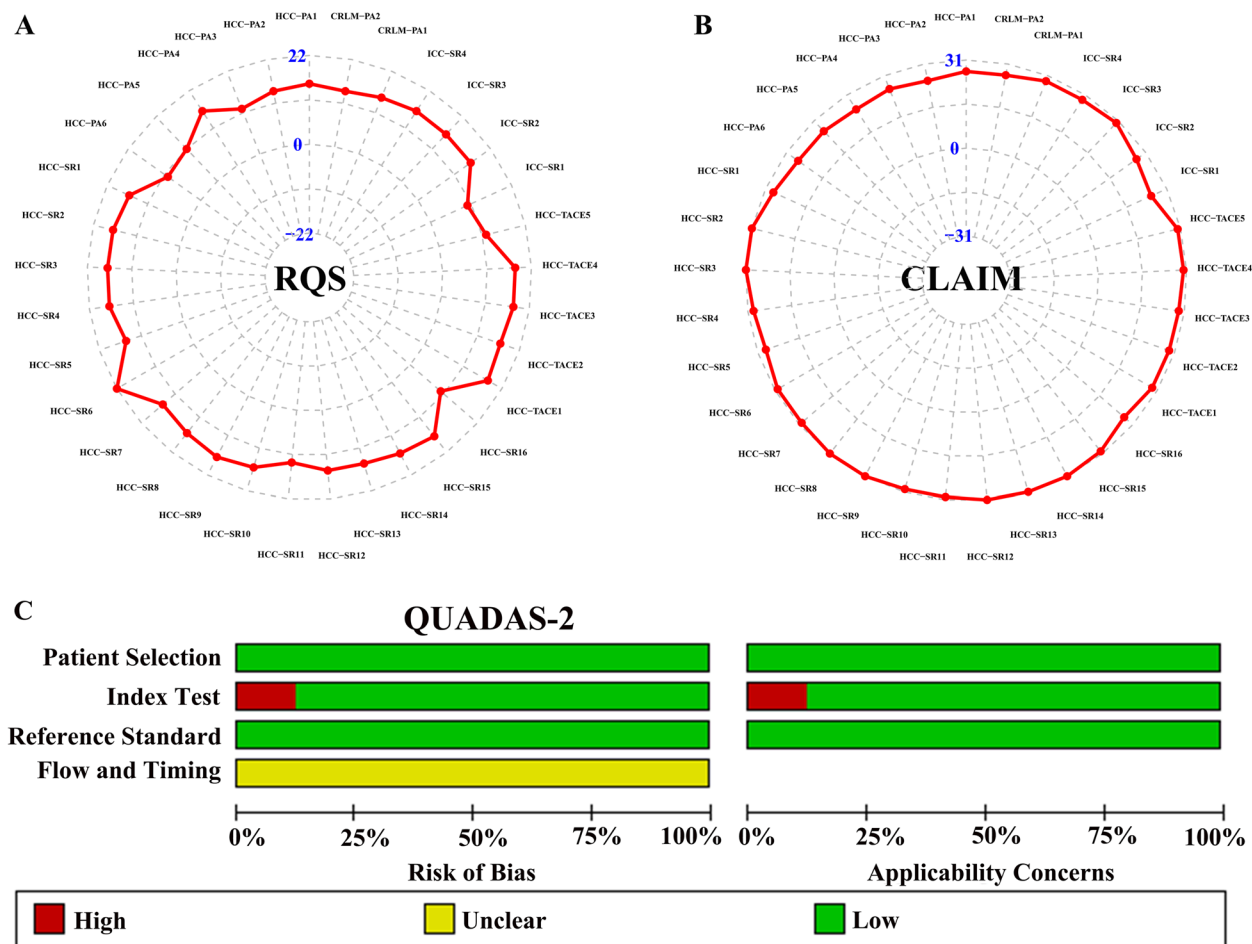


Fig. 2 Quality assessment. **(A)** RQS. The RQS results showed a median total score of 16, ranging from 10 to 22. **(B)** CLAIM. The median CLAIM score was 71% (30/42). **(C)** QUADAS-2. 29 studies were deemed to have a low risk of bias

34 cohorts with 3379 patients treated by SR, and 10 cohorts with 1,148 patients treated by TACE.

The meta-analysis based on PA with AI, clinical, combined models revealed that the overall SENS and SPEC were 0.78 (95% confidence Interval [CI], 0.69–0.85; $I^2=58$, $P=0.01$), 0.90 (95% CI, 0.86–0.92; $I^2=54$, $P=0.03$); 0.78 (95% CI, 0.70–0.85; $I^2=60$, $P=0.02$), 0.68 (95% CI, 0.54–0.79; $I^2=91$, $P<0.01$); 0.87 (95% CI, 0.80–0.92; $I^2=64$, $P<0.01$) 0.84 (95% CI, 0.75–0.90; $I^2=84$, $P<0.01$), respectively. In the Cochrane diagnostic, the comprehensive analyses with high I^2 values ($>50\%$) and statistical significance indicate strong heterogeneity in the overall SENS and SPEC analyses of all models. The sROC curves demonstrated an overall AUC of 0.92 for AI models, 0.81 for clinical models, and 0.92 for combined models. The presence of publication bias was detected by Deeks of funnel plot asymmetry test, and the results showed no publication bias (AI models, $P=0.92$; clinical models, $P=0.49$; combined models, $P=0.70$).

The meta-analysis based on SR with AI, clinical, combined models revealed that the overall SENS and SPEC were 0.81 (95% CI, 0.77–0.84; $I^2=65$, $P<0.01$), 0.77 (95% CI, 0.73–0.81; $I^2=72$, $P<0.01$); 0.66 (95% CI, 0.61–0.71; $I^2=67$, $P<0.01$), 0.80 (95% CI, 0.75–0.83; $I^2=54$, $P<0.01$); 0.83 (95% CI, 0.79–0.86; $I^2=71$, $P<0.01$), 0.81 (95% CI, 0.78–0.84; $I^2=55$, $P<0.01$), respectively. In the Cochrane diagnostic, the comprehensive analyses with high I^2 values ($>50\%$) and statistical significance indicate strong heterogeneity in the overall SENS and SPEC analyses of all models. The sROC curves demonstrated an overall AUC of 0.86 for AI models, 0.80 for clinical models, and 0.89 for combined models. The presence of publication bias was detected by Deeks of funnel plot asymmetry test, and the results showed no publication bias (AI models, $P=0.27$; clinical models, $P=0.57$; combined models, $P=0.50$).

The meta-analysis based on TACE with AI, clinical, combined models revealed that the overall SENS and SPEC were 0.73 (95% CI, 0.68–0.77; $I^2=0$, $P=0.71$),

Table 2 2 × 2 data table for predicting recurrence of liver cancer after first-line treatment based on AI

Author	Year	Population	TP	FP	FN	TN	Treatment
Ma QP et al.-T	2021	255	51	16	25	163	HCC-PA
Ma QP et al.-V	2021	63	11	2	6	44	HCC-PA
Chen C et al.-T	2023	189	15	12	4	158	HCC-PA
Chen C et al.-V1	2023	135	11	18	1	105	HCC-PA
Chen C et al.-V2	2023	93	11	7	2	73	HCC-PA
Li M et al.-T	2022	91	53	2	5	31	HCC-PA
Li M et al.-V	2022	40	16	2	10	12	HCC-PA
Li MH et al.-T	2022	112	26	16	5	65	HCC-PA
Li MH et al.-V	2022	28	5	4	3	16	HCC-PA
Gao W et al.-T	2022	378	95	84	12	187	HCC-SR
Gao W et al.-V	2022	94	19	16	7	52	HCC-SR
Kang W et al.-T	2023	112	52	10	13	37	HCC-SR
Kang W et al.-V	2023	48	23	3	9	13	HCC-SR
Wang J et al.-T	2022	180	86	19	29	47	HCC-SR
Wang J et al.-V	2022	28	14	3	4	7	HCC-SR
Wang Q et al.-T	2023	83	22	1	20	40	HCC-SR
Wang Q et al.-V	2023	36	15	6	3	12	HCC-SR
Zhao Y et al.-T	2021	78	34	10	6	28	HCC-SR
Zhao Y et al.-V	2021	35	13	5	5	12	HCC-SR
Zhang Z et al.-T	2019	108	48	31	4	25	HCC-SR
Zhang Z et al.-V	2019	47	16	7	7	17	HCC-SR
Zhang H et al.-T	2022	121	43	21	5	52	HCC-SR
Zhang H et al.-V	2022	51	16	12	4	19	HCC-SR
Yan M et al.-T	2023	195	41	8	13	133	HCC-SR
Yan M et al.-V	2023	90	18	7	5	60	HCC-SR
Wu C et al.-T	2022	91	34	11	10	36	HCC-SR
Wu C et al.-V	2022	41	16	4	4	17	HCC-SR
Wang L et al.-T	2022	133	45	22	11	55	HCC-SR
Wang L et al.-V	2022	57	21	13	3	20	HCC-SR
Ning P et al.-T	2020	225	87	17	34	87	HCC-SR
Ning P et al.-V	2020	100	33	11	22	34	HCC-SR
Lv C et al.-T	2023	156	69	14	8	65	HCC-SR
Lv C et al.-V	2023	68	25	8	13	22	HCC-SR
Li W et al.-T	2022	211	85	36	9	81	HCC-SR
Li W et al.-V	2022	91	43	13	4	31	HCC-SR
Li ZY et al.-T	2023	108	48	11	7	42	HCC-SR
Li ZY et al.-V	2023	45	23	3	5	14	HCC-SR
Wang L et al.-T	2023	77	14	12	2	49	HCC-SR
Wang L et al.-V	2023	33	7	4	1	21	HCC-SR
Chen C et al.-T	2023	138	42	19	13	64	HCC-TACE
Chen C et al.-V	2023	35	11	4	3	17	HCC-TACE
Peng G et al.-T	2024	248	60	20	23	145	HCC-TACE
Peng G et al.-V	2024	107	24	19	11	53	HCC-TACE
Jin Z-T et al.-T	2021	136	27	23	7	79	HCC-TACE
Jin Z-T et al.-V	2021	120	19	27	7	67	HCC-TACE
Wang DD et al.-T	2022	171	38	14	20	99	HCC-TACE
Wang DD et al.-V	2022	72	14	8	9	41	HCC-TACE
Kong XL et al.-T	2021	84	32	11	13	28	HCC-TACE
Kong XL et al.-V	2021	37	17	7	3	10	HCC-TACE

Table 2 (continued)

Author	Year	Population	TP	FP	FN	TN	Treatment
Jolissaint JS et al.-T	2022	97	23	10	5	59	ICC-SR
Jolissaint JS et al.-V	2022	41	7	7	4	23	ICC-SR
Zhu Y et al.-T	2021	92	22	24	2	44	ICC-SR
Zhu Y et al.-V	2021	33	10	9	1	13	ICC-SR
Liang W et al.-T	2018	139	66	13	19	41	ICC-SR
Liang W et al.-V	2018	70	45	11	3	11	ICC-SR
Zhao QX et al.-T	2023	153	45	13	29	66	CRLM-PA
Zhao QX et al.-V1	2023	32	15	7	1	9	CRLM-PA
Zhao QX et al.-T2	2023	22	7	4	2	9	CRLM-PA
Hu H et al.-T	2023	315	46	105	20	144	CRLM-PA
Hu H et al.-V	2023	140	22	42	10	66	CRLM-PA

0.79 (95% CI, 0.74–0.83; $I^2=63$, $P<0.01$); 0.64 (95% CI, 0.53–0.75; $I^2=46$, $P=0.13$), 0.78 (95% CI, 0.68–0.86; $I^2=80$, $P<0.01$); 0.82 (95% CI, 0.75–0.87; $I^2=16$, $P=0.31$), 0.82 (95% CI, 0.76–0.86; $I^2=42$, $P=0.13$), respectively. In the Cochrane diagnostic, the comprehensive analyses with high I^2 values ($>50\%$) and statistical significance indicate strong heterogeneity in the overall SROC analyses of AI and clinical models. The sROC curves demonstrated an overall AUC of 0.79 for AI models, 0.77 for clinical models, and 0.89 for combined models. The presence of publication bias was detected by Deeks of funnel plot asymmetry test, and the results showed no publication bias (AI models, $P=0.10$; clinical models, $P=0.65$; combined models, $P=0.23$).

The meta-analysis based on PA, SR, TACE with AI, clinical, combined models revealed that the overall SENS and SPEC were 0.79 (95% CI, 0.75–0.81; $I^2=59$, $P<0.01$), 0.80 (95% CI, 0.77–0.83; $I^2=77$, $P<0.01$); 0.69 (95% CI, 0.64–0.73; $I^2=67$, $P<0.01$), 0.77 (95% CI, 0.73–0.81; $I^2=81$, $P<0.01$); 0.84 (95% CI, 0.80–0.86; $I^2=66$, $P<0.01$) 0.83 (95% CI, 0.80–0.85; $I^2=65$, $P<0.01$), respectively. In the Cochrane diagnostic, the comprehensive analyses with high I^2 values ($>50\%$) and statistical significance indicate strong heterogeneity in the overall SENS and SPEC analyses of all models. The sROC curves demonstrated an overall AUC of 0.86 for AI models, 0.79 for clinical models, and 0.90 for combined models. The presence of publication bias was detected by Deeks of funnel plot asymmetry test, and the results showed no publication bias (AI models, $P=0.49$; clinical models, $P=0.79$; combined models, $P=0.40$). The comprehensive performance based on AI were displayed in Figs. 3 and 4.

Comprehensive performance of recurrence prediction after first-line treatment of ICC based on artificial intelligence

The baseline characteristics in the meta-analysis of ICC recurrence prediction were presented in Table 2. One of the SR studies only entered a meta-analysis of combined models [75]. A total of 8 cohorts with 681 patients treated by SR were analyzed. The meta-analysis based on AI models, clinical models, combined models revealed that the overall SENS and SPEC were 0.85 (95% CI, 0.76–0.91; $I^2=53$, $P=0.06$), 0.71 (95% CI, 0.60–0.80; $I^2=68$, $P=0.01$); 0.72 (95% CI, 0.65–0.78; $I^2=0$, $P=0.93$), 0.66 (95% CI, 0.60–0.73; $I^2=35$, $P=0.15$); 0.90 (95% CI, 0.83–0.94; $I^2=70$, $P<0.01$), 0.84 (95% CI, 0.75–0.90; $I^2=81$, $P<0.01$). In the Cochrane diagnostic test, comprehensive analyses with high I^2 values ($>50\%$) and statistical significance indicate strong heterogeneity in the overall sensitivity and specificity analyses of AI and combined models, while clinical models exhibit lower heterogeneity. The sROC curves demonstrated an overall AUC of 0.86 for AI models; 0.75 for clinical models; 0.94 for combined models. The presence of publication bias was detected by Deeks of funnel plot asymmetry test, and the results showed no publication bias (AI models, $P=0.76$; clinical models, $P=0.52$; combined model, $P=0.35$).

Comprehensive performance of recurrence prediction after first-line treatment of CLRM based on artificial intelligence

The baseline characteristics in the meta-analysis of CLRM recurrence prediction were presented in Table 2. A total of 5 cohorts with 662 patients treated by PA were analyzed. The meta-analysis based on AI models, clinical models, combined models revealed that the overall

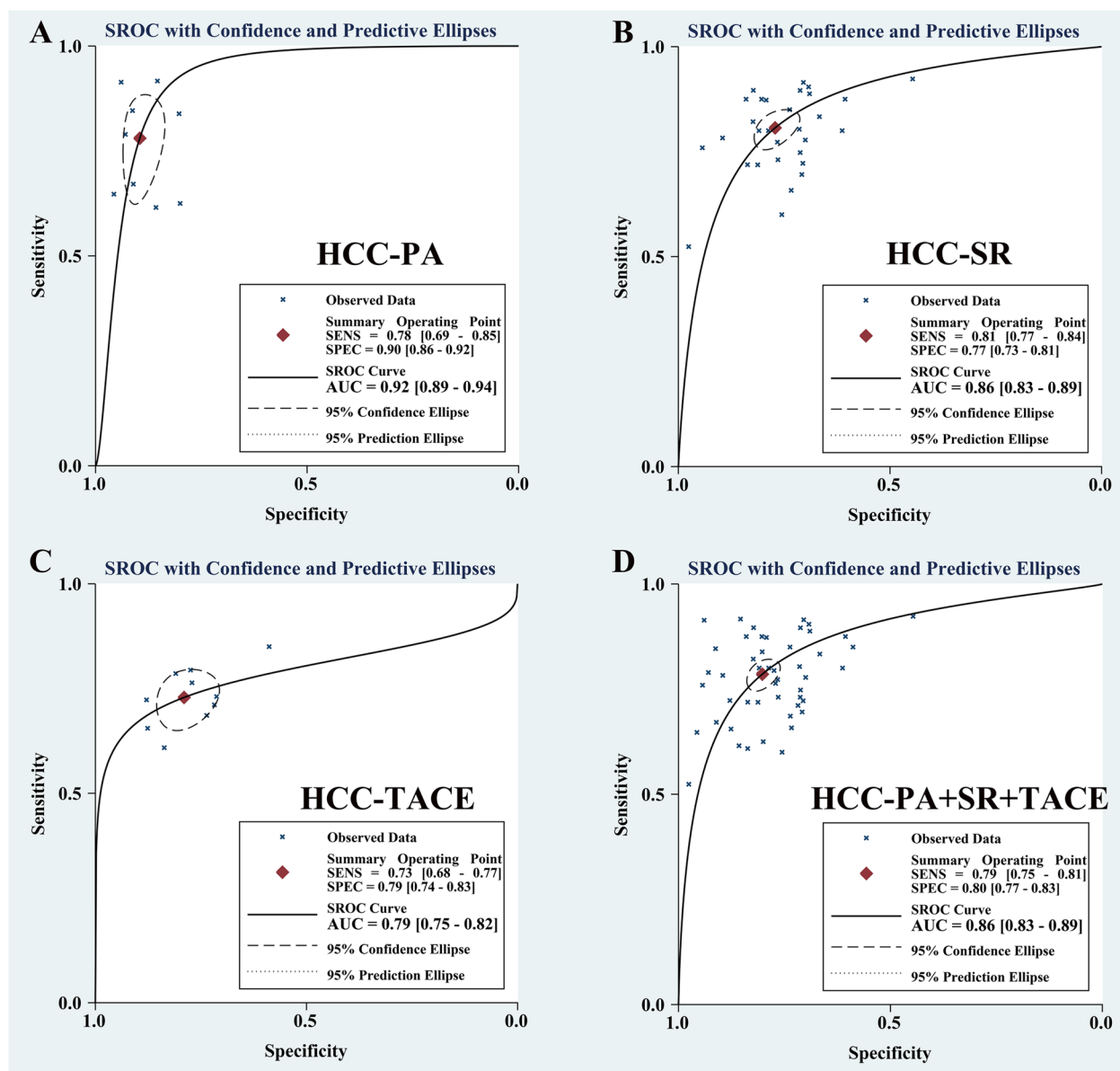


Fig. 3 Comprehensive performance of recurrence prediction after first-line treatment of HCC based on AI models. (A) PA. (B) SR. (C) TACE. (D) PA, SR, and TACE. The sROC curves demonstrated an overall AUC of 0.92 for PA, 0.86 for SR, 0.79 for TACE, 0.86 for PA, SR and TACE

SENS and SPEC were 0.69 (95% CI, 0.62–0.75; $I^2=53$, $P=0.07$), 0.63 (95% CI, 0.59–0.68; $I^2=80$, $P<0.01$); 0.63 (95% CI, 0.56–0.70; $I^2=90$, $P<0.01$), 0.70 (95% CI, 0.68–0.74; $I^2=0$, $P=0.71$); 0.82 (95% CI, 0.76–0.82; $I^2=48$, $P=0.11$), 0.86 (95% CI, 0.83–0.89; $I^2=0$, $P=0.51$). In the Cochrane diagnostic test, comprehensive analyses with high I^2 values ($>50\%$) and statistical significance indicate strong heterogeneity in the overall sensitivity and specificity analyses of AI models, while clinical and combined models exhibit lower heterogeneity. The sROC curves demonstrated an overall AUC of 0.74 for AI models; 0.75

for clinical models; 0.92 for combined models (Table 3). The presence of publication bias was detected by Deeks of funnel plot asymmetry test, and the results showed no publication bias (AI models, $P=0.30$; clinical models, $P=0.69$; combined model, $P=0.37$).

Clinical feature and subgroup analysis

In the combined model, tumor number, tumor size, alpha-fetoprotein, and microvascular invasion were the clinical features that frequently occurred (Table 4), but

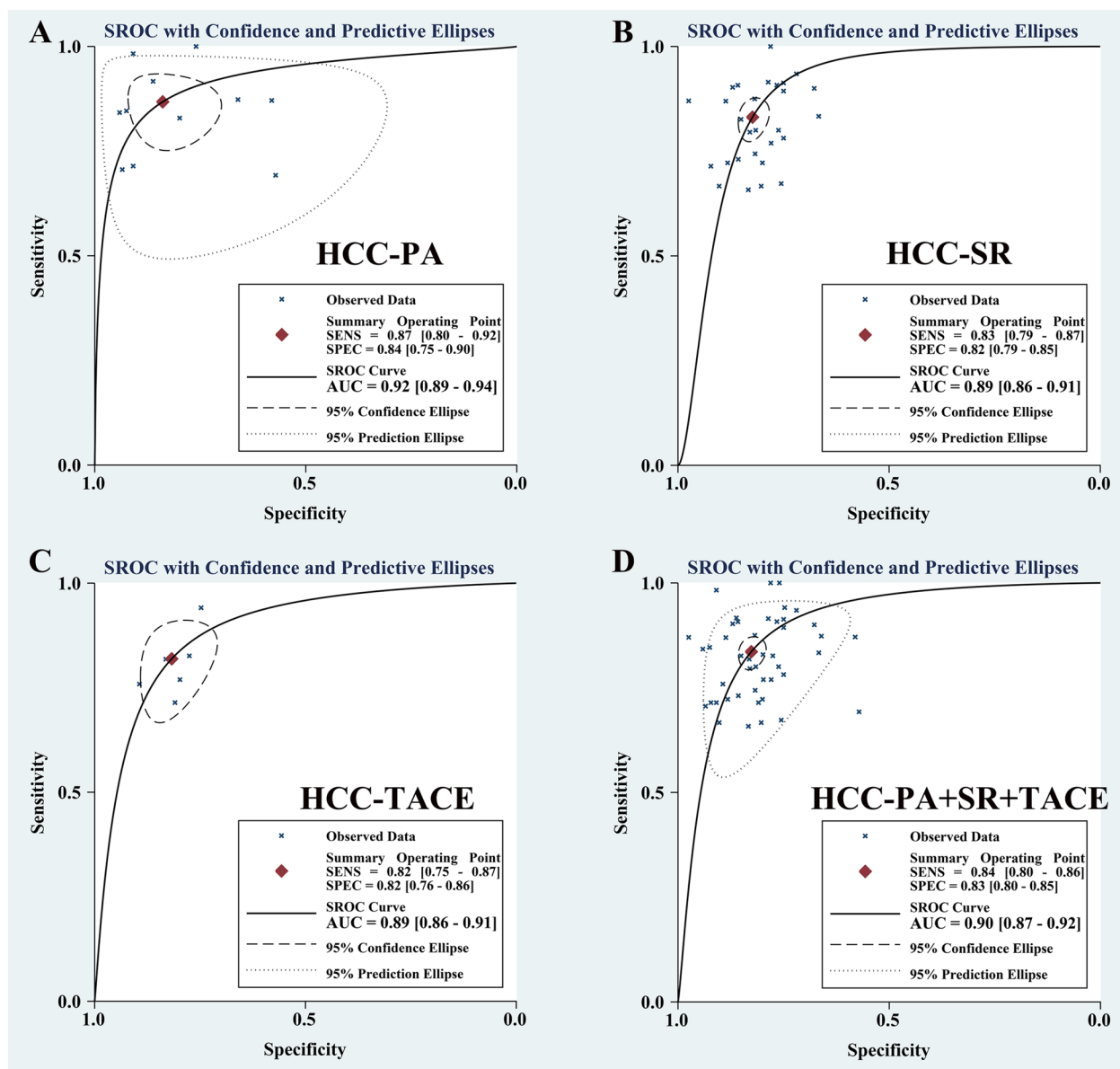


Fig. 4 Comprehensive performance of recurrence prediction after first-line treatment of HCC based on combined models. (A) PA. (B) SR. (C) TACE. (D) PA, SR, and TACE. The sROC curves demonstrated an overall AUC of 0.92 for PA, 0.89 for SR, 0.89 for TACE, 0.90 for PA, SR and TACE

the number and type of clinically significant features varied in each study, which was due to many confounding factors in observational studies.

Based on the recurrence time, image modality, AI method, and cohort type, subgroup analyses were conducted and summarized in Table 5. In the AI subgroup analysis for HCC-PA, the pooled AUCs for recurrence time > 2 years, CEUS, deep learning, and the training cohort were superior to those with recurrence time ≤ 2 years, MRI, radiomics, and the validation cohort,

respectively. For the AI subgroup analyses of HCC-SR and HCC-TACE, the recurrence time, AI approach, and cohort type were similar to those in the HCC-PA analysis, with a slightly better pooled AUC based on MRI compared to CT image modality. In the AI subgroup analysis for ICC-SR, the pooled AUC of the training cohort was superior to that of the validation cohort. The AI analysis for CRLM was not suitable for subgroup analysis due to the limited number of included studies.

Table 3 Comprehensive performance of artificial intelligence in predicting recurrence after first-Line treatment of liver cancer

Liver cancer	Study number	Models	SENC	SPEC	AUC
HCC-PA	4	AI	0.78 (0.69–0.85)	0.90 (0.86–0.92)	0.92
	3	Clinical	0.78 (0.70–0.85)	0.68 (0.54–0.79)	0.81
	4	Combined	0.86 (0.79–0.91)	0.84 (0.74–0.91)	0.92
HCC-SR	15	AI	0.81 (0.77–0.84)	0.77 (0.73–0.81)	0.86
	10	Clinical	0.66 (0.61–0.71)	0.80 (0.75–0.83)	0.80
	14	Combined	0.84 (0.79–0.87)	0.82 (0.80–0.85)	0.89
HCC-TACE	5	AI	0.73 (0.68–0.77)	0.79 (0.74–0.83)	0.79
	3	Clinical	0.64 (0.53–0.75)	0.78 (0.68–0.86)	0.77
	3	Combined	0.82 (0.75–0.87)	0.82 (0.76–0.86)	0.89
ICC-SR	3	AI	0.85 (0.75–0.91)	0.71 (0.60–0.80)	0.86
	3	Clinical	0.72 (0.65–0.77)	0.66 (0.59–0.72)	0.75
	4	Combined	0.90 (0.83–0.94)	0.84 (0.75–0.90)	0.94
CRLM-PA	2	AI	0.69 (0.62–0.75)	0.63 (0.59–0.68)	0.74
	2	Clinical	0.63 (0.56–0.70)	0.70 (0.68–0.74)	0.75
	2	Combined	0.82 (0.76–0.82)	0.86 (0.83–0.89)	0.92

Table 4 Clinical features for combined AI models

Study	Year	Clinical features
Lv X et al	2021	Tumor shape, ADC value, DWI signal intensity, ΔSI
Ma QP et al.-T	2021	APRI, platelet, monocytes, US_radiomics
Yang X et al.-T	2022	Tumor number, integrity of the capsule, biopsy result, log10AFP
Chen C et al.-T	2023	Tumor number, ablative margin, location abutting major vessels
Li M et al.-T	2022	Tumor size, alpha-fetoprotein
Gao W et al.-T	2022	Rad_score
Kang W et al.-T	2023	Alpha-fetoprotein, microvascular invasion
Wang J et al.-T	2022	Hepatitis B surface antigen, gamma-glutamyl transferase, alpha-fetoprotein
Wang Q et al.-T	2023	Pathologic differentiation grade, tumor number, intratumoral necrosis measured with MRI, infiltrative margin, wash-out measured with CT
Zhao Y et al.-T	2021	Microvascular invasion, pathological grading, tumor size,
Zhang Z et al.-T	2019	Alpha-fetoprotein, gross vascular invasion, non-smooth tumour margin
Zhang H et al.-T	2022	Rad_score
Yan M et al.-T	2023	Neutrophil count, aspartate aminotransferase, microvascular invasion
Wu C et al.-T	2022	Edmondson grade
Wang L et al.-T	2022	Arterial peritumoral enhancement, non-smooth tumor margin, satellite nodules, cirrhosis, serosal invasion, albumin
Ning P et al.-T	2020	Tumor capsule, tumor size, gamma-glutamyl transferase
Lv C et al.-T	2023	Barcelona clinic liver cancer staging, microvascular invasion, α -fetoprotein, DL_feature
Li W et al.-T	2022	Two-trait predictor of venous invasion, rim enhancement, tumor capsule
Li J et al.-T	2024	Alpha-fetoprotein, microvascular invasion
Chen C et al.-T	2023	Spleen_radiomics score
Jin Z-T et al.-T	2021	Serum bilirubin level, tumor number, tumor capsule
Wang DD et al.-T	2022	Tumor size, first response, alpha-fetoprotein, tumor enhancement
Jolissaint JS et al.-T	2022	Tumor size on imaging
Zhu Y et al.-T	2021	Solitary, membrane invasion
Song Y et al.-T	2023	Carbohydrate antigen 19–9, vascular invasion, tumor margin
Liang W et al.-T	2018	TNM stage
Zhao QX et al.-T	2023	Lymph node metastasis of the primary colorectal cancer, preoperative chemotherapy, advanced stage
Hu H et al.-T	2023	Adjuvant chemotherapy before liver microwave ablation, maximum tumor diameter, ablative margin, TNM stage of primary tumor

Table 5 Subgroup analysis

Therapy	Factor	Cohort number	AUC	SENC	SPEC
HCC-PA	> 2 year	4	0.96	0.74	0.92
	≤ 2 year	5	0.91	0.82	0.88
	CEUS	4	0.96	0.74	0.92
	MRI	5	0.91	0.82	0.88
	RAD	4	0.92	0.81	0.84
	DL	5	0.94	0.72	0.91
	training cohort	4	0.93	0.79	0.9
	validation cohort	5	0.91	0.71	0.88
HCC-SR	> 2 year	20	0.87	0.81	0.77
	≤ 2 year	10	0.83	0.77	0.73
	CEUS	14	0.85	0.77	0.79
	MRI	14	0.88	0.83	0.74
	RAD	20	0.85	0.79	0.74
	DL	10	0.87	0.81	0.77
	training cohort	15	0.88	0.81	0.76
	validation cohort	15	0.82	0.76	0.75
HCC-TACE	≤ 2 year	4	0.82	0.76	0.74
	CT	6	0.83	0.72	0.79
	MRI	4	0.81	0.73	0.81
	training cohort	5	0.83	0.72	0.83
ICC-SR	validation cohort	5	0.79	0.72	0.74
	CT	4	0.86	0.84	0.74
	training cohort	3	0.87	0.81	0.75
	validation cohort	3	0.79	0.89	0.64

The application of artificial intelligence-based meta-analysis in liver cancer

In the PubMed database, three AI meta-analyses on HCC recurrence were found, while no AI meta-analysis on ICC or CRLM recurrence was identified. These three meta-analyses included 49, 15, and 10 studies respectively, none of which provided a detailed analysis of treatment protocols, which encompassed single or combined therapy [91–93]. However, one of the meta-analyses provided a detailed summary of AI's prediction of HCC recurrence based on cohort and image type, revealing

combined AUCs of 0.79 and 0.75 for CT, 0.83 and 0.79 for MRI, and 0.87 and 0.76 for CEUS in the training and validation cohorts. The other two meta-analyses reported combined AUCs of 0.83 and 0.89, respectively (Table 6).

Discussion

The management of liver cancer, including HCC, ICC, and CRLM, typically involves first-line treatments such as SR, PA, and TACE. However, post-treatment recurrence remains a significant clinical challenge, necessitating the exploration of combined therapeutic approaches, such as targeted or immunotherapy, for patients experiencing relapse. Accurately identifying patients prone to liver cancer recurrence and subsequently delivering precision post-surgical interventions remains a critical area of research. AI methods have been extensively applied to predict recurrence following first-line treatments for liver cancer. The effectiveness of AI in predicting recurrence after treatment for primary liver cancer was satisfactory, with moderate to high efficacy. However, in CRLM, AI still need further exploration to improve the prediction of recurrence after treatment. Despite the use of multiple quality assessment methods, the meta-analysis still exhibited considerable heterogeneity, a finding consistent with many other AI meta-analyses.

The progress of artificial intelligence in primary malignant focal liver lesions

Through the survey of multiple databases, it had been revealed that over 300 original studies had been conducted on the application of AI in predicting recurrence after liver cancer treatment, primarily focusing on first-line therapy for HCC, including PA, SR, or TACE. However, regarding ICC, no AI studies on recurrence prediction after PA and TACE treatments had been identified, with only a dozen studies found on recurrence prediction after SR treatment. For CRLM, over a dozen studies had been discovered investigating AI in recurrence prediction after SR or PA treatment. This aligned with the global statistics on liver cancer burden. Additionally, the application of AI in predicting recurrence after liver transplantation had also been investigated,

Table 6 The application of artificial intelligence-based meta-analysis in liver cancer

Author	Year	Study number	Image type	Cohort number	AUC	SENC	SPEC
Jin J et al.-T	2024	11	CT	11	-	0.79 (0.70–0.85)	0.82 (0.75–0.87)
Jin J et al.-V	2024	11	CT	11	-	0.72 (0.59–0.83)	0.77 (0.68–0.84)
Jin J et al.-T	2024	10	MRI	10	-	0.84 (0.76–0.89)	0.77 (0.66–0.85)
Jin J et al.-V	2024	10	MRI	11	-	0.74 (0.68–0.79)	0.78 (0.70–0.84)
Tian H et al	2023	15	CT, MRI	-	0.83 (0.79–0.86)	0.75 (0.65–0.82)	0.78 (0.68–0.85)
Lu M et al	2024	10	MRI	-	0.89 (0.86–0.91)	0.84 (0.78–0.89)	0.80 (0.75–0.85)

yet only one study meeting the criteria for meta-analysis was found. This study utilized CT images from a single-center cohort of 151 HCC patients to develop a machine learning model, achieving a SENC and SPEC of 0.86 and 0.85 for predicting recurrence after liver transplantation [94]. The AI methods employed include radiomics and deep learning, primarily utilizing images from CT, MRI, and US. Interestingly, pathomics based on quantitative analysis of postoperative pathological images was gradually being explored. For instance, a deep learning analysis of whole-slide images from multiple centers for HCC had achieved an AUC of 0.72 in predicting recurrence after SR treatment [95]. Furthermore, a recurrence-related histological score constructed using deep learning methods based on 576 whole-slide images predicted recurrence with an AUC of 0.74 [96]. The findings suggest that further exploration is still needed in applying AI for recurrence prediction after ICC and CLRM treatments. Additionally, pathomics, as a quantitative analysis based on pathological images, is still far from meeting the requirements of precision medicine.

Subsequently, the application of meta-analyses in liver cancer was investigated. In the PubMed database, three meta-analyses utilizing AI for HCC recurrence prediction were identified, while no AI meta-analysis on ICC or CRLM recurrence was found. These three meta-analyses encompassed 49, 15, and 10 studies, respectively, none of which provided detailed analysis on the treatment regimen, including single or combination therapy. However, one of the meta-analyses offered a detailed summary of AI's predictive capabilities for HCC recurrence based on cohort and image type. From the above, it was evident that published meta-analyses had not focused on the value of AI in predicting recurrence after a single first-line treatment. However, considering the variability in recurrence rates among different first-line treatment options, such as PA treatment leading to a higher recurrence rate compared to SR, overlooking the treatment regimen in recurrence assessment may overestimate the accuracy of AI in recurrence prediction. Therefore, this review, unlike other meta-analyses, would focus on the application of AI in predicting recurrence after first-line treatment for liver cancer, aiming to assess the value of AI in each first-line treatment regimen, and precisely identify high-risk populations for recurrence and facilitate timely addition of combination therapy.

Quality assessment

This review employed three quality assessment methods: RQS, QUADAS-2, and CLAIM. Among them, RQS and QUADAS-2 were commonly used in numerous meta-analyses. The average RQS score in this review was 16, ranging from 10 to 22, similar to the findings of other

meta-analyses [91–93], indicating a moderate-to-low quality. The primary reasons for the relatively low RQS quality were associated with factors such as prospective, cost, open science, phantom study, multiple timepoints, and biological correlates, which had been disclosed in numerous meta-analyses, raising questions about their applicability. QUADAS-2 deemed 29 studies to have a low risk of bias. However, it was often observed that there were biases and application issues related to index testing, primarily due to the lack of detailed reports on image preprocessing. QUADAS-2 seems to overestimate methodological quality, which is also similar to other meta-analyses [97]. CLAIM, a quality assessment method introduced in recent years, encompasses 42 quality control items, overlapping with RQS to a great extent, but expanding to quality control of various sections of the research writing. CLAIM has gradually been adopted in the quality assessment of AI meta-analyses. For instance, in a meta-analysis evaluating COVID-19 pneumonia versus other pneumonias, 19 studies using deep learning methods had an average CLAIM score of 20, with the highest score being 29 and the lowest 14 [98]. In another meta-analysis evaluating the prediction of medulloblastoma subtypes, 5 radiomics studies achieved an average CLAIM score of 23 [99]. In this review, the median CLAIM score was 71% (30/42). Some CLAIM items were typically reported, including abstract structure, study introduction, study goal, inclusion criteria for the study population, model performance metrics, performance metrics to distinguish the optimal model(s), and study limitations. Few studies disclosed factors such as de-identification methods, how missing data were handled, intended sample size and its determination, initialization of model parameters, failure analysis of incorrectly classified cases, registration number and name of registry, and where the full study protocol could be accessed. RQS highlighted the specific issue of phantom studies in radiomics, while CLAIM pointed out deficiencies in missing data handling, sample size or power calculations, as well as shortcomings in data de-identification and failure analysis [24]. Both RQS and CLAIM emphasized the importance of data robustness, which was intimately tied to segmentation methods. These methods encompass manual, semi-automatic, and fully automatic approaches, leveraging techniques such as gray-level based methods, learning-based techniques, and graph-cut methods. Segmentation methods could exhibit variability depending on the application scenario and imaging modality. Manual segmentation, a commonly employed technique in radiomics, relied on the assessment of intra-observer and inter-observer variability to evaluate robustness. For instance, the reproducibility of features among observers could quantitatively assess the magnetic susceptibility of

intracranial dissecting aneurysms, as evidenced by the study [100]. Employing learning-based techniques, such as deep learning, could lead to pseudo-segmentation if the training data selection was inadequate, as deep learning heavily relied on the quality and representativeness of the training data. While fully automatic segmentation eliminates human factors, it too could result in pseudo-segmentation, yet it remained a promising avenue for further development in machine learning applications. This was particularly evident in the context of deep learning for ultrasound images, where manual segmentation of thousands of images was impractical, highlighting the advantages of fully automatic approaches [101, 102]. In addition to segmentation techniques, imaging artifacts, including noise, motion, and others, significantly impact the performance of segmentation methods, as exemplified in [103]. Currently, no single segmentation method can yield acceptable results for every medical image, making the selection of an appropriate method for a given segmentation task a challenging dilemma. Currently, no single segmentation method could yield acceptable results for every medical image, making the selection of an appropriate method for a given segmentation task a challenging dilemma [104].

Comprehensive predictive performance

This review incorporated six PA studies, 16 SR studies, and five TACE studies to predict the recurrence of HCC after treatment. In the evaluation of recurrence after PA and SR treatment, the comprehensive assessment performance was satisfactory, achieving moderate to high performance with AUCs of 0.92 and 0.86, respectively. However, in the assessment of recurrence after TACE treatment, the comprehensive SENC, SPEC, and AUC were 0.73, 0.79, and 0.79, respectively, indicating only moderate performance. These results were similar to the conclusions of published meta-analyses on HCC recurrence prediction [91–93]. However, the AI meta-analysis based on PA treatment recurrence showed a higher conclusion compared to other meta-analyses, which was likely related to the differences in recurrence among first-line treatment options mentioned in this review, and could also be attributed to the relatively small number of studies included. In the prediction of recurrence after ICC-SR and CLRM-PA treatment, the comprehensive AUCs were 0.86 and 0.74, respectively. It was evident that AI demonstrates more stable performance in predicting recurrence after SR treatment, which may be associated with the more extensive application of SR in liver cancer. Furthermore, by combining clinical pathological features, AI had improved the performance of predicting recurrence after first-line treatment to varying degrees. This review, as same as the previous meta-analysis, analyzed

the total predictive performance of AI in all first-line treatment schemes, and AUC reached 0.86, with similar performance. However, it underestimated the high predictive performance of AI in the recurrence of PA treatment, and overestimated the low predictive performance of AI in the recurrence of TACE treatment.

Limitations

This review encountered several noteworthy limitations: (1) The number of studies included in the recurrence evaluation of ICC and CLRM were relatively small, which could not completely exclude the possibility of over-exclusion during the literature screening process. Therefore, the generalizability of the meta-analysis results remained a relevant topic for discussion. (2) The meta-analysis results revealed a high degree of heterogeneity in SENS and SPEC. This heterogeneity was primarily attributed to the lack of standardization in scientific rigor and clinical relevance. (3) AI studies had limitations, such as the robustness of manual segmentation and model over-fitting, which still require more standardized quality control protocols to improve study quality [105]. (4) There were limitations in the evaluation of RQS, QUADAS-2, and CLAIM, and the evaluation criteria were interpreted based on relevant literature in this study, which was still needed to more effectively evaluate the quality of AI. (5) All the studies were retrospective, lacking validation of the effectiveness of AI in predicting HCC recurrence through high-quality, multi-center, prospective studies. (6) The studies included in this review were mainly from China, and the global applicability of the conclusions still required regional validation.

Conclusions

This systematic review and meta-analysis demonstrates the comprehensive application value of AI in predicting recurrence after a single first-line liver cancer treatment, with satisfactory results indicating the clinical translation potential of AI in predicting recurrence. However, it must be acknowledged that due to the complexity of the AI analysis process, there are variations in studies that lead to heterogeneity in the meta-analysis. Therefore, further development and improvement of quality assessment standards for AI methodologies are still needed to achieve the goal of precise evaluation through the generalized application of AI.

Abbreviations

AI	Artificial intelligence
AUC	The area under the curve
SENC	Sensitivity
SPEC	Specificity
PA	Percutaneous ablation
SR	Surgical resection
TACE	Transarterial chemoembolization

HCC	Hepatocellular carcinoma
ICC	Intrahepatic cholangiocarcinoma
CRLM	Colorectal cancer liver metastasis
ER	Early recurrence
CEA	Carcinoembryonic antigen
CA19-9	Carbohydrate antigen 19–9
CT	Computed tomography
MRI	Magnetic resonance imaging
US	Ultrasound
CEUS	Contrast-enhanced ultrasound
TP	True positive
FN	False negative
FP	False positive
TN	True negative
RQS	The Radiomics Quality Score
CLAIM	Checklist for Artificial Intelligence in Medical Imaging
QUADAS-2	Quality Assessment of Diagnostic Accuracy Studies
sROC	The summary receiver operating characteristic

Supplementary Information

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Supplementary Material 1.
Supplementary Material 2.
Supplementary Material 3.
Supplementary Material 4.
Supplementary Material 5.

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Authors' contributions

Data curation, L.Y., D.Y., and Q.F.; Formal analysis, L.Y. and Q.F.; Funding acquisition, L.Y.; Investigation, L.Y., Q.F., and S.H.; Methodology, Y.Z., H.J., S.F., and Q.L.; Project administration, L.Y., D.Y., and Q.F.; Resources, L.Y., D.Y., and Q.F.; Software, L.Y.; Supervision, Y.Z., H.J., S.F., and Q.L.; Validation, D.Y. and Q.F.; Writing – original draft, L.Y.; Writing – review & editing, D.Y., Q.F., Y.Z., H.J., S.F., and Q.L. All authors reviewed the manuscript.

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Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Rumgay H, Ferlay J, de Martel C, et al. Global, regional and national burden of primary liver cancer by subtype. *Eur J Cancer*. 2022;161:108–18. <https://doi.org/10.1016/j.ejca.2021.11.023>.
- Horn SR, Stoltzfus KC, Lehrer EJ, et al. Epidemiology of liver metastases. *Cancer Epidemiol*. 2020;67:101760. <https://doi.org/10.1016/j.canep.2020.101760>.
- Shin SW, Ahn KS, Kim SW, et al. Liver Resection Versus Local Ablation Therapies for Hepatocellular Carcinoma Within the Milan Criteria: A Systematic Review and Meta-analysis. *Ann Surg*. 2021;273:656–66. <https://doi.org/10.1097/SLA.0000000000004350>.
- Minami Y, Aoki T, Hagiwara S, et al. Tips for Preparing and Practicing Thermal Ablation Therapy of Hepatocellular Carcinoma. *Cancers (Basel)*. 2023;15:4763. <https://doi.org/10.3390/cancers15194763>.
- Chakraborty E, Sarkar D. Emerging Therapies for Hepatocellular Carcinoma (HCC). *Cancers (Basel)*. 2022;14:2798. <https://doi.org/10.3390/cancers14112798>.
- Dhote A, Tzedakis S, Shapira OI, et al. Current status and perspectives in the surgical and oncological management of intrahepatic cholangiocarcinoma. *J Visc Surg*. 2023;160:346–55. <https://doi.org/10.1016/j.jvisurg.2023.07.007>.
- Masoud SJ, Rhodin KE, Kanu E, et al. Comparing Survival After Resection, Ablation, and Radiation in Small Intrahepatic Cholangiocarcinoma. *Ann Surg Oncol*. 2023;30:6639–46. <https://doi.org/10.1245/s10434-023-13872-y>.
- Itenberg ER, Lozano AM. Surgical and Interventional Management of Liver Metastasis. *Clin Colon Rectal Surg*. 2022;37:80–4. <https://doi.org/10.1055/s-0042-1758822>.
- Tawara S, Miyazaki T, Kiyota R, et al. Comparison of percutaneous ablation and hepatectomy for liver metastasis: A single center retrospective study. *Cancer Med*. 2024;13: e6957. <https://doi.org/10.1002/cam4.6957>.
- Wu Z, Zeng Y, Yuan Y, et al. Early recurrence of hepatocellular carcinoma in patients after ablation and resection: A propensity score analysis. *Am J Surg*. 2024;228:94–101. <https://doi.org/10.1016/j.amjsurg.2023.08.002>.
- Wang C, Pang S, Si-Ma H, et al. Specific risk factors contributing to early and late recurrences of intrahepatic cholangiocarcinoma after curative resection. *World J Surg Oncol*. 2019;17:2. <https://doi.org/10.1186/s12957-018-1540-1>.
- Zhao QX, He XL, Wang K, et al. Deep learning model based on contrast-enhanced ultrasound for predicting early recurrence after thermal ablation of colorectal cancer liver metastasis. *Eur Radiol*. 2023;33:1895–905. <https://doi.org/10.1007/s00330-022-09203-6>.
- Liu L, Qin S, Lin K, et al. Development and comprehensive validation of a predictive prognosis model for very early HCC recurrence within one year after curative resection: a multicenter cohort study. *Int J Surg*. 2024. <https://doi.org/10.1097/JS9.0000000000001467>.
- Alaimo L, Lima HA, Moazzam Z, et al. Development and Validation of a Machine-Learning Model to Predict Early Recurrence of Intrahepatic Cholangiocarcinoma. *Ann Surg Oncol*. 2023;30:5406–15. <https://doi.org/10.1245/s10434-023-13636-8>.
- Dai S, Ye Y, Kong X, et al. A predictive model for early recurrence of colorectal-cancer liver metastases based on clinical parameters. *Gastroenterol Rep (Oxf)*. 2021;9:241–51. <https://doi.org/10.1093/gastro/goaa092>.
- Choi WJ, Williams PJ, Claassen MP, et al. Systematic Review and Meta-Analysis of Prognostic Factors for Early Recurrence in Intrahepatic Cholangiocarcinoma After Curative-Intent Resection. *Ann Surg Oncol*. 2022. <https://doi.org/10.1245/s10434-022-11463-x>.
- Tian Y, Wang Y, Wen N, et al. Prognostic factors associated with early recurrence following liver resection for colorectal liver metastases: a systematic review and meta-analysis. *BMC Cancer*. 2024;24:426. <https://doi.org/10.1186/s12885-024-12162-4>.
- Ansari MY, Abdalla A, Ansari MY, et al. Practical utility of liver segmentation methods in clinical surgeries and interventions. *BMC Med Imaging*. 2022;22:97. <https://doi.org/10.1186/s12880-022-00825-2>.

19. Rai P, Ansari MY, Warfa M, et al. Efficacy of fusion imaging for immediate post-ablation assessment of malignant liver neoplasms: A systematic review. *Cancer Med*. 2023;12:14225–51. <https://doi.org/10.1002/cam4.6089>.
20. Ansari MY, Qaraqe M, Righetti R, et al. Unveiling the future of breast cancer assessment: a critical review on generative adversarial networks in elastography ultrasound. *Front Oncol*. 2023;13:1282536. <https://doi.org/10.3389/fonc.2023.1282536>.
21. Han, Z, Jian, M, Wang, G. ConvUNet: An efficient convolution neural network for medical image segmentation. *KNOWL-BASED SYST*. 2022; 109512. <https://doi.org/10.1016/j.knosys.2022.109512>
22. Ma QP, He XL, Li K, et al. Dynamic Contrast-Enhanced Ultrasound Radiomics for Hepatocellular Carcinoma Recurrence Prediction After Thermal Ablation. *Mol Imaging Biol*. 2021;23:572–85. <https://doi.org/10.1007/s11307-021-01578-0>.
23. Jolissaint JS, Wang T, Soares KC, et al. Machine learning radiomics can predict early liver recurrence after resection of intrahepatic cholangiocarcinoma. *HPB (Oxford)*. 2022;24:1341–50. <https://doi.org/10.1016/j.hpb.2022.02.004>.
24. Zhong J, Hu Y, Zhang G, et al. An updated systematic review of radiomics in osteosarcoma: utilizing CLAIM to adapt the increasing trend of deep learning application in radiomics. *Insights Imaging*. 2022;13:138. <https://doi.org/10.1186/s13244-022-01277-6>.
25. Sollini M, Antunovic L, Chiti A, Kirienco M. Towards clinical application of image mining: a systematic review on artificial intelligence and radiomics. *Eur J Nucl Med Mol Imaging*. 2019;46:2656–72. <https://doi.org/10.1007/s00259-019-04372-x>.
26. Hectors SJ, Lewis S, Besa C, et al. MRI radiomics features predict immuno-oncological characteristics of hepatocellular carcinoma. *Eur Radiol*. 2020;30:3759–69. <https://doi.org/10.1007/s00330-020-06675-2>.
27. Liu Z, Liu Y, Zhang W, et al. Deep learning for prediction of hepatocellular carcinoma recurrence after resection or liver transplantation: a discovery and validation study. *Hepatol Int*. 2022;16:577–89. <https://doi.org/10.1007/s12072-022-10321-y>.
28. Wen L, Weng S, Yan C, et al. A Radiomics Nomogram for Preoperative Prediction of Early Recurrence of Small Hepatocellular Carcinoma After Surgical Resection or Radiofrequency Ablation. *Front Oncol*. 2021;11:657039. <https://doi.org/10.3389/fonc.2021.657039>.
29. Shen JX, Zhou Q, Chen ZH, et al. Longitudinal radiomics algorithm of posttreatment computed tomography images for early detecting recurrence of hepatocellular carcinoma after resection or ablation. *Transl Oncol*. 2021;14:100866. <https://doi.org/10.1016/j.tranon.2020.100866>.
30. Shan QY, Hu HT, Feng ST, et al. CT-based peritumoral radiomics signatures to predict early recurrence in hepatocellular carcinoma after curative tumor resection or ablation. *Cancer Imaging*. 2019;19:11. <https://doi.org/10.1186/s40644-019-0197-5>.
31. Liu F, Liu D, Wang K, et al. Deep Learning Radiomics Based on Contrast-Enhanced Ultrasound Might Optimize Curative Treatments for Very-Early or Early-Stage Hepatocellular Carcinoma Patients. *Liver Cancer*. 2020;9:397–413. <https://doi.org/10.1159/000505694>.
32. Kucukkaya AS, Zeevi T, Chai NX, et al. Predicting tumor recurrence on baseline MR imaging in patients with early-stage hepatocellular carcinoma using deep machine learning. *Sci Rep*. 2023;13:7579. <https://doi.org/10.1038/s41598-023-34439-7>.
33. Della Corte A, Mori M, Calabrese F, et al. Preoperative MRI radiomic analysis for predicting local tumor progression in colorectal liver metastases before microwave ablation. *Int J Hyperthermia*. 2024;41:2349059. <https://doi.org/10.1080/02656736.2024.2349059>.
34. Xie Q, Zhao Z, Yang Y, et al. Radiomics-guided prognostic assessment of early-stage hepatocellular carcinoma recurrence post-radical resection. *J Cancer Res Clin Oncol*. 2023;149:14983–96. <https://doi.org/10.1007/s00432-023-05291-z>.
35. Zhou Y, He L, Huang Y, et al. CT-based radiomics signature: a potential biomarker for preoperative prediction of early recurrence in hepatocellular carcinoma. *Abdom Radiol (NY)*. 2017;42:1695–704. <https://doi.org/10.1007/s00261-017-1072-0>.
36. Shahveranova A, Balli HT, Aikimbaev K, et al. Prediction of Local Tumor Progression After Microwave Ablation in Colorectal Carcinoma Liver Metastases Patients by MRI Radiomics and Clinical Characteristics-Based Combined Model: Preliminary Results. *Cardiovasc Intervent Radiol*. 2023;46:713–25. <https://doi.org/10.1007/s00270-023-03454-6>.
37. Xue X, Li B, Feng K, et al. Clinical value of enhanced CT texture analysis technique in predicting short-term recurrence of liver tumor in patients with HCC after TACE. *J Prac Hepatol*. 2020;23:423–6.
38. Dong X, Gao J, Su W, et al. Texture analysis based on preoperative MRI to predict liver cancer early recurrence after transarterial chemoembolization. *J Chin Onco*. 2023;29:140–4.
39. Wang L, Feng B, Wang S, et al. Diagnostic value of whole-tumor apparent diffusion coefficient map radiomics analysis in predicting early recurrence of solitary hepatocellular carcinoma ≤ 5 cm. *Abdom Radiol (NY)*. 2022;47:3290–300. <https://doi.org/10.1007/s00261-022-03582-6>.
40. Lv X, Chen M, Kong C, et al. Construction of a novel radiomics nomogram for the prediction of aggressive intrasegmental recurrence of HCC after radiofrequency ablation. *Eur J Radiol*. 2021;144:109955. <https://doi.org/10.1016/j.ejrad.2021.109955>.
41. Yang X, Yuan C, Zhang Y, et al. Predicting hepatocellular carcinoma early recurrence after ablation based on magnetic resonance imaging radiomics nomogram. *Medicine (Baltimore)*. 2022;101:e32584. <https://doi.org/10.1097/MD.00000000000032584>.
42. Chen C, Han Q, Ren H, et al. Multiparametric MRI-based model for prediction of local progression of hepatocellular carcinoma after thermal ablation. *Cancer Med*. 2023;12:17529–40. <https://doi.org/10.1002/cam4.6277>.
43. Li M, Zeng N, Zhou Y, et al. Prediction of early recurrence risk of hepatocellular carcinoma after ablation by preoperative contrast-enhanced ultrasound omics model. *Mod Digestion Intervent*. 2022;27:20–5+30.
44. Li W, Xu C, Cui J, et al. The value of predicting early recurrence of hepatocellular carcinoma after radiofrequency ablation based on pre-treatment MR-T2WI radiomics features. *J Med Information*. 2022;35:39–43+52.
45. Gao W, Wang W, Song D, et al. A predictive model integrating deep and radiomics features based on gadobenate dimeglumine-enhanced MRI for postoperative early recurrence of hepatocellular carcinoma. *Radiol Med*. 2022;127:259–71. <https://doi.org/10.1007/s11547-021-01445-6>.
46. Kang W, Cao X, Luo J. Effect of multiple peritumoral regions of interest ranges based on computed tomography radiomics for the prediction of early recurrence of hepatocellular carcinoma after resection. *Quant Imaging Med Surg*. 2023;13:6668–82. <https://doi.org/10.21037/qims-23-226>.
47. Wang J, Wu D, Sun M, et al. Deep Segmentation Feature-Based Radiomics Improves Recurrence Prediction of Hepatocellular Carcinoma. *BME Front*. 2022;2022:9793716. <https://doi.org/10.34133/2022/9793716>.
48. Wang Q, Sheng Y, Jiang Z, et al. What Imaging Modality Is More Effective in Predicting Early Recurrence of Hepatocellular Carcinoma after Hepatectomy Using Radiomics Analysis: CT or MRI or Both? *Diagnostics (Basel)*. 2023;13:2012. <https://doi.org/10.3390/diagnostics13122012>.
49. Zhao Y, Wu J, Zhang Q, et al. Radiomics Analysis Based on Multiparametric MRI for Predicting Early Recurrence in Hepatocellular Carcinoma After Partial Hepatectomy. *J Magn Reson Imaging*. 2021;53:1066–79. <https://doi.org/10.1002/jmri.27424>.
50. Zhang Z, Jiang H, Chen J, et al. Hepatocellular carcinoma: radiomics nomogram on gadoteric acid-enhanced MR imaging for early postoperative recurrence prediction. *Cancer Imaging*. 2019;19:22. <https://doi.org/10.1186/s40644-019-0209-5>.
51. Zhang H, Huo F. Prediction of early recurrence of HCC after hepatectomy by contrast-enhanced ultrasound-based deep learning radiomics. *Front Oncol*. 2022;12:930458. <https://doi.org/10.3389/fonc.2022.930458>.
52. Yan M, Zhang X, Zhang B, et al. Deep learning nomogram based on Gd-EOB-DTPA MRI for predicting early recurrence in hepatocellular carcinoma after hepatectomy. *Eur Radiol*. 2023;33:4949–61. <https://doi.org/10.1007/s00330-023-09419-0>.
53. Wu C, Yu S, Zhang Y, et al. CT-Based Radiomics Nomogram Improves Risk Stratification and Prediction of Early Recurrence in Hepatocellular Carcinoma After Partial Hepatectomy. *Front Oncol*. 2022;12:896002. <https://doi.org/10.3389/fonc.2022.896002>.
54. Wang L, Ma X, Feng B, et al. Multi-Sequence MR-Based Radiomics Signature for Predicting Early Recurrence in Solitary Hepatocellular Carcinoma ≤ 5 cm. *Front Oncol*. 2022;12:899404. <https://doi.org/10.3389/fonc.2022.899404>.

55. Ning P, Gao F, Hai J, et al. Application of CT radiomics in prediction of early recurrence in hepatocellular carcinoma. *Abdom Radiol (NY)*. 2020;45:64–72. <https://doi.org/10.1007/s00261-019-02198-7>.
56. Lv C, He N, Yang JJ, et al. Prediction of 3-year recurrence rate of hepatocellular carcinoma after resection based on contrast-enhanced CT: a single-centre study. *Br J Radiol*. 2023;96:20220702. <https://doi.org/10.1259/bjr.20220702>.
57. Li W, Shen H, Han L, et al. A Multiparametric Fusion Radiomics Signature Based on Contrast-Enhanced MRI for Predicting Early Recurrence of Hepatocellular Carcinoma. *J Oncol*. 2022;2022:3704987. <https://doi.org/10.1155/2022/3704987>.
58. Li J, Ma Y, Yang C, et al. Radiomics analysis of R2* maps to predict early recurrence of single hepatocellular carcinoma after hepatectomy. *Front Oncol*. 2024;14:1277698. <https://doi.org/10.3389/fonc.2024.1277698>.
59. Wang L, Zhang W, Niu S. Predictive value of CT radiomics characteristics for early recurrence after hepatocellular carcinoma surgery. *Prac J Can*. 2023;38:1510–4.
60. Li Z, Yu J, Li Yue, et al. Radiomics model based on CT predicts early recurrence after surgical resection of CNLC stage I-II hepatocellular carcinoma. *J Clinical Radiology*. 2023;42:1590–1596. <https://doi.org/10.13437/j.cnki.jcr.2023.10.027>.
61. Chen C, Liu J, Gu Z, et al. Integration of Multimodal Computed Tomography Radiomic Features of Primary Tumors and the Spleen to Predict Early Recurrence in Patients with Postoperative Adjuvant Transarterial Chemoembolization. *J Hepatocell Carcinoma*. 2023;10:1295–308. <https://doi.org/10.2147/JHC.S423129>.
62. Peng G, Cao X, Huang X, et al. Radiomics and machine learning based on preoperative MRI for predicting extrahepatic metastasis in hepatocellular carcinoma patients treated with transarterial chemoembolization. *Eur J Radiol Open*. 2024;12:100551. <https://doi.org/10.1016/j.ejro.2024.100551>.
63. Jin Z, Chen L, Zhong B, et al. Machine-learning analysis of contrast-enhanced computed tomography radiomics predicts patients with hepatocellular carcinoma who are unsuitable for initial transarterial chemoembolization monotherapy: A multicenter study. *Transl Oncol*. 2021;14:101034. <https://doi.org/10.1016/j.tranon.2021.101034>.
64. Wang DD, Zhang JF, Zhang LH, et al. Clinical-radiomics predictors to identify the suitability of transarterial chemoembolization treatment in intermediate-stage hepatocellular carcinoma: A multicenter study. *Hepatobiliary Pancreat Dis Int*. 2023;22:594–604. <https://doi.org/10.1016/j.hbpd.2022.11.005>.
65. Sun X, You Y, Zhao X, et al. Prediction of the early recurrence of HCC patients after TACE surgery based on T2 weighted image texture analysis. *Chin J Magn Reson Imaging*. 2021;12:22–6+32.
66. Fiz F, Masci C, Costa G, et al. PET/CT-based radiomics of mass-forming intrahepatic cholangiocarcinoma improves prediction of pathology data and survival. *Eur J Nucl Med Mol Imaging*. 2022;49:3387–400. <https://doi.org/10.1007/s00259-022-05765-1>.
67. Zhu Y, Mao Y, Chen J, et al. Radiomics-based model for predicting early recurrence of intrahepatic mass-forming cholangiocarcinoma after curative tumor resection. *Sci Rep*. 2021;11:18347. <https://doi.org/10.1038/s41598-021-97796-1>.
68. Wakiya T, Ishido K, Kimura N, et al. CT-based deep learning enables early postoperative recurrence prediction for intrahepatic cholangiocarcinoma. *Sci Rep*. 2022;12:8428. <https://doi.org/10.1038/s41598-022-12604-8>.
69. Hao X, Liu B, Hu X, et al. A Radiomics-based Approach for Predicting Early Recurrence in Intrahepatic Cholangiocarcinoma after Surgical Resection: A Multicenter Study. *Annu Int Conf IEEE Eng Med Biol Soc*. 2021;2021:3659–62. <https://doi.org/10.1109/EMBC46164.2021.9630029>.
70. Chu H, Liu Z, Liang W, et al. Radiomics using CT images for preoperative prediction of futile resection in intrahepatic cholangiocarcinoma. *Eur Radiol*. 2021;31:2368–76. <https://doi.org/10.1007/s00330-020-07250-5>.
71. Chen B, Mao Y, Li J, et al. Predicting very early recurrence in intrahepatic cholangiocarcinoma after curative hepatectomy using machine learning radiomics based on CECT: A multi-institutional study. *Comput Biol Med*. 2023;167:107612. <https://doi.org/10.1016/j.compbiomed.2023.107612>.
72. Park HJ, Park B, Park SY, et al. Preoperative prediction of postsurgical outcomes in mass-forming intrahepatic cholangiocarcinoma based on clinical, radiologic, and radiomics features. *Eur Radiol*. 2021;31:8638–48. <https://doi.org/10.1007/s00330-021-07926-6>.
73. Song Y, Zhou G, Zhou Y, et al. Artificial intelligence CT radiomics to predict early recurrence of intrahepatic cholangiocarcinoma: a multi-center study. *Hepatol Int*. 2023;17:1016–27. <https://doi.org/10.1007/s12072-023-10487-z>.
74. Xu L, Wan Y, Luo C, et al. Integrating intratumoral and peritumoral features to predict tumor recurrence in intrahepatic cholangiocarcinoma. *Phys Med Biol*. 2021;66:https://doi.org/10.1088/1361-6560/ac01f3. <https://doi.org/10.1088/1361-6560/ac01f3>.
75. Liang W, Xu L, Yang P, et al. Novel Nomogram for Preoperative Prediction of Early Recurrence in Intrahepatic Cholangiocarcinoma. *Front Oncol*. 2018;8:360. <https://doi.org/10.3389/fonc.2018.00360>.
76. Taghavi M, Staal F, Gomez Munoz F, et al. CT-Based Radiomics Analysis Before Thermal Ablation to Predict Local Tumor Progression for Colorectal Liver Metastases. *Cardiovasc Intervent Radiol*. 2021;44:913–20. <https://doi.org/10.1007/s00270-020-02735-8>.
77. van der Reijdt DJ, Guendel C, Staal FCR, et al. Independent validation of CT radiomics models in colorectal liver metastases: predicting local tumour progression after ablation. *Eur Radiol*. 2023. <https://doi.org/10.1007/s00330-023-10417-5>.
78. Staal FCR, Taghavi M, van der Reijdt DJ, et al. Predicting local tumour progression after ablation for colorectal liver metastases: CT-based radiomics of the ablation zone. *Eur J Radiol*. 2021;141:109773. <https://doi.org/10.1016/j.ejrad.2021.109773>.
79. Taghavi M, Staal FC, Simões R, et al. CT radiomics models are unable to predict new liver metastasis after successful thermal ablation of colorectal liver metastases. *Acta Radiol*. 2023;64:5–12. <https://doi.org/10.1177/02841851211060437>.
80. Hu H, Chi JC, Zhai B, et al. CT-based radiomics analysis to predict local progression of recurrent colorectal liver metastases after microwave ablation. *Medicine (Baltimore)*. 2023;102:e36586. <https://doi.org/10.1097/MD.00000000000036586>.
81. Granata V, Fusco R, De Muzio F, et al. Contrast MR-Based Radiomics and Machine Learning Analysis to Assess Clinical Outcomes following Liver Resection in Colorectal Liver Metastases: A Preliminary Study. *Cancers (Basel)*. 2022;14:1110. <https://doi.org/10.3390/cancers14051110>.
82. Saber R, Henault D, Messaoudi N, et al. Radiomics using computed tomography to predict CD73 expression and prognosis of colorectal cancer liver metastases. *J Transl Med*. 2023;21:507. <https://doi.org/10.1186/s12967-023-04175-7>.
83. Granata V, Fusco R, Setola SV, et al. CT-Based Radiomics Analysis to Predict Histopathological Outcomes Following Liver Resection in Colorectal Liver Metastases. *Cancers (Basel)*. 2022;14:1648. <https://doi.org/10.3390/cancers14071648>.
84. Granata V, Fusco R, De Muzio F, et al. EOB-MR Based Radiomics Analysis to Assess Clinical Outcomes following Liver Resection in Colorectal Liver Metastases. *Cancers (Basel)*. 2022;14:1239. <https://doi.org/10.3390/cancers14051239>.
85. Granata V, Fusco R, De Muzio F, et al. Radiomics textural features by MR imaging to assess clinical outcomes following liver resection in colorectal liver metastases. *Radiol Med*. 2022;127:461–70. <https://doi.org/10.1007/s11547-022-01477-6>.
86. Wang Q, Nilsson H, Xu K, et al. Exploring tumor heterogeneity in colorectal liver metastases by imaging: Unsupervised machine learning of preoperative CT radiomics features for prognostic stratification. *Eur J Radiol*. 2024;175:111459. <https://doi.org/10.1016/j.ejrad.2024.111459>.
87. Shur J, Orton M, Connor A, et al. A clinical-radiomic model for improved prognostication of surgical candidates with colorectal liver metastases. *J Surg Oncol*. 2020;121:357–64. <https://doi.org/10.1002/jso.25783>.
88. Tang L, Zhang Z, Yang J, et al. A New Automated Prognostic Prediction Method Based on Multi-Sequence Magnetic Resonance Imaging for Hepatic Resection of Colorectal Cancer Liver Metastases. *IEEE J Biomed Health Inform*. 2024;28:1528–39. <https://doi.org/10.1109/JBHI.2024.3350247>.
89. Chen J, Cheung HMC, Karanickolas PJ, et al. A radiomic biomarker for prognosis of resected colorectal cancer liver metastases generalizes across MRI contrast agents. *Front Oncol*. 2023;13:898854. <https://doi.org/10.3389/fonc.2023.898854>.
90. Kobe A, Zraggen J, Messmer F, et al. Prediction of treatment response to transarterial radioembolization of liver metastases: Radiomics analysis of pre-treatment cone-beam CT: A proof of concept study. *Eur J Radiol Open*. 2021;8:100375. <https://doi.org/10.1016/j.ejro.2021.100375>.

91. Jin J, Jiang Y, Zhao YL, et al. Radiomics-based Machine Learning to Predict the Recurrence of Hepatocellular Carcinoma: A Systematic Review and Meta-analysis. *Acad Radiol*. 2024;31:467–79. <https://doi.org/10.1016/j.acra.2023.09.008>.
92. Tian H, Xie Y, Wang Z. Radiomics for preoperative prediction of early recurrence in hepatocellular carcinoma: a meta-analysis. *Front Oncol*. 2023;13:1114983. <https://doi.org/10.3389/fonc.2023.1114983>.
93. Lu M, Wang C, Zhuo Y, et al. Preoperative prediction power of radiomics and non-radiomics methods based on MRI for early recurrence in hepatocellular carcinoma: a systemic review and meta-analysis. *Abdom Radiol (NY)*. <https://doi.org/10.1007/s00261-024-04356-y>
94. Zhao JW, Shu X, Chen XX, et al. Prediction of early recurrence of hepatocellular carcinoma after liver transplantation based on computed tomography radiomics nomogram. *Hepatobiliary Pancreat Dis Int*. 2022;21:543–50. <https://doi.org/10.1016/j.hbpd.2022.05.013>.
95. Yamashita R, Long J, Saleem A, Rubin DL, Shen J. Deep learning predicts postsurgical recurrence of hepatocellular carcinoma from digital histopathologic images. *Sci Rep*. 2021;11:2047. <https://doi.org/10.1038/s41598-021-81506-y>.
96. Qu WF, Tian MX, Qiu JT, et al. Exploring pathological signatures for predicting the recurrence of early-stage hepatocellular carcinoma based on deep learning. *Front Oncol*. 2022;12:968202. <https://doi.org/10.3389/fonc.2022.968202>.
97. Liu L, Liao H, Zhao Y, et al. CT-based radiomics for predicting lymph node metastasis in esophageal cancer: a systematic review and meta-analysis. *Front Oncol*. 2024;14:1267596. <https://doi.org/10.3389/fonc.2024.1267596>.
98. Jia LL, Zhao JX, Pan NN, et al. Artificial intelligence model on chest imaging to diagnose COVID-19 and other pneumonias: A systematic review and meta-analysis. *Eur J Radiol Open*. 2022;9:100438. <https://doi.org/10.1016/j.ejro.2022.100438>.
99. Karabacak M, Ozkara BB, Ozturk A, et al. Radiomics-based machine learning models for prediction of medulloblastoma subgroups: a systematic review and meta-analysis of the diagnostic test performance. *Acta Radiol*. 2023;64:1994–2003. <https://doi.org/10.1177/02841851221143496>.
100. Park SI, Kim D, Jung SC, et al. Feasibility and intra-and interobserver reproducibility of quantitative susceptibility mapping with radiomic features for intracranial dissecting intramural hematomas and atherosclerotic calcifications. *Sci Rep*. 2023;13:3651. <https://doi.org/10.1038/s41598-023-30745-2>.
101. Al-Kababji A, Bensaali F, Dakua S, et al. Automated liver tissues delineation techniques: A systematic survey on machine learning current trends and future orientations. *ENG APPL ARTIF INTEL*. 2023;117:105532. <https://doi.org/10.1016/j.engappai.2022.105532>.
102. Zhai X, Eslami M, Hussein E, et al. Real-time automated image segmentation technique for cerebral aneurysm on reconfigurable system-on-chip J COMPUT SCI-NETH. 2018;27:35–45. <https://doi.org/10.1016/j.jjocs.2018.05.002>.
103. Mohanty S, Dakua S. Toward Computing Cross-Modality Symmetric Non-Rigid Medical Image Registration. *IEEE Access*. 2022;10:24528–39. <https://doi.org/10.1109/access.2022.3154771>.
104. Dakua S. Performance divergence with data discrepancy: a review. *ARTIF INTELL REV*. 2011;40:429–55. <https://doi.org/10.1007/s10462-011-9289-8>.
105. Akhtar, Y, Dakua, S, Abdalla, A, et al. Risk Assessment of Computer-aided Diagnostic Software for Hepatic Resection *IEEE T RADIAT PLASMA*. 2021;1–1. <https://doi.org/10.1109/trpms.2021.3071148>

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